WORKED SOLUTION

STEP I 2007, Question 4

Pure • Common Roots of Quadratics

STEP I 2007, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s1-q4

The key idea

The first part is the classic factorisation of $x^3 + y^3 + z^3 - 3xyz$, and once you know $x + b + c$ is a factor the quadratic follows by matching coefficients. In the second part we have two quadratics sharing a root. The standard trick is to *eliminate*: combine the two equations so that one power of k cancels. Doing this in two different ways gives k and k^2 as ratios, and then $k^2 = (k)^2$ produces a relation between the coefficients alone. That relation turns out to be the first part with $x = a$.

Solution

The factorisation

Put $x = -(b + c)$ into $P(x) = x^3 - 3xbc + b^3 + c^3$:

$$\begin{aligned} P(-b-c) &= -(b+c)^3 + 3bc(b+c) + b^3 + c^3 \\ &= -(b^3 + 3b^2c + 3bc^2 + c^3) + 3b^2c + 3bc^2 + b^3 + c^3 = 0, \end{aligned}$$

so by the factor theorem $x + b + c$ is a factor. Matching coefficients (the x^2 term of P is zero, and the constant term is $b^3 + c^3$) gives

$$x^3 - 3xbc + b^3 + c^3 = (x + b + c)\,\mathrm{Q}(x), \qquad \mathrm{Q}(x) = x^2 + b^2 + c^2 - xb - bc - cx.$$

Expanding the right-hand side confirms it: the x^2 terms are $x^2(b + c) - x^2(b + c) = 0$, the x terms collect to $-3bcx$, and the constants are $b^3 + c^3$.

For the squares, each of x^2, b^2, c^2 appears twice in $2\mathrm{Q}(x)$ and each cross term appears as $-2xb$ and so on, so

$$2\mathrm{Q}(x) = (x - b)^2 + (b - c)^2 + (c - x)^2.$$

The common root

Since k is a root of both equations,

$$ak^2 + bk + c = 0, \tag{1}$$
$$bk^2 + ck + a = 0. \tag{2}$$

To remove k^2, take $b \times (1) - a \times (2)$:

$$(b^2 - ac)k + bc - a^2 = 0, \qquad \text{so} \qquad (ac - b^2)\,k = bc - a^2.$$

To remove the k terms instead, take $c \times (1) - b \times (2)$:

$$(ac - b^2)k^2 + c^2 - ab = 0, \qquad \text{so} \qquad (ac - b^2)\,k^2 = ab - c^2.$$

This is the similar expression involving k^2.

Now use $k^2 = k \cdot k$. Since $ac \neq b^2$ we may divide by $ac - b^2$, giving $k = \dfrac{bc - a^2}{ac - b^2}$, and so

$$(ac - b^2)\,\frac{(bc - a^2)^2}{(ac - b^2)^2} = ab - c^2 \quad \Longrightarrow \quad (ac - b^2)(ab - c^2) = (bc - a^2)^2,$$

after multiplying through by $ac - b^2$. This is the required result.

Expanding both sides,

$$a^2bc - ac^3 - ab^3 + b^2c^2 = b^2c^2 - 2a^2bc + a^4,$$
$$\text{so} \quad 0 = a^4 + ab^3 + ac^3 - 3a^2bc = a\left(a^3 - 3abc + b^3 + c^3\right).$$

Since $a \neq 0$, we get $a^3 - 3abc + b^3 + c^3 = 0$, as required.

Ask yourself Where have we seen $a^3 - 3abc + b^3 + c^3$ before?

It is the first part with $x = a$. So

$$(a + b + c)\,\mathrm{Q}(a) = 0, \qquad 2\mathrm{Q}(a) = (a - b)^2 + (b - c)^2 + (c - a)^2.$$

Hence either $a + b + c = 0$ or $\mathrm{Q}(a) = 0$.

Case $\mathrm{Q}(a) = 0$**.** Because a, b, c are real, the three squares are non-negative, and their sum is zero only if each is zero. So $a = b = c$, and the two equations are both $a(y^2 + y + 1) = 0$: they are identical.

Case $a + b + c = 0$**.** Substitute $c = -a - b$ into our formula for k:

$$bc - a^2 = -ab - b^2 - a^2, \qquad ac - b^2 = -a^2 - ab - b^2,$$

so $bc - a^2 = ac - b^2$, and since this is non-zero, $k = 1$. (Indeed $y = 1$ satisfies both equations, as each then reads $a + b + c = 0$.)

So either $k = 1$ or the two equations are identical, which completes the question.

Notice that in the identical case $a = b = c$ gives $ac = b^2$, which the hypotheses rule out. So under the stated conditions the common root is in fact always $k = 1$; the "identical" alternative is what happens if you drop the condition $ac \neq b^2$, and then the common roots are the complex cube roots of unity.

Checks

Take $a = 1$, $b = 2$, $c = -3$, so $a + b + c = 0$. The equations are $y^2 + 2y - 3 = (y - 1)(y + 3) = 0$ and $2y^2 - 3y + 1 = (2y - 1)(y - 1) = 0$, which share $k = 1$. Here $ac - b^2 = -7$, $bc - a^2 = -7$ and $ab - c^2 = -7$, so $(ac - b^2)k = bc - a^2$, $(ac - b^2)k^2 = ab - c^2$ and $(-7)(-7) = (-7)^2$ all hold, and $a^3 - 3abc + b^3 + c^3 = 1 + 18 + 8 - 27 = 0$. We also tested both identities of the first part, and the expansion $(ac - b^2)(ab - c^2) - (bc - a^2)^2 = -a(a^3 - 3abc + b^3 + c^3)$, in exact arithmetic on a thousand random triples.

Common slips

- Eliminating the constant term $(a \times (1) - c \times (2))$ gives $k\big((a^2 - bc)k + ab - c^2\big) = 0$, which still needs $k \neq 0$ to be useful. The combination $c \times (1) - b \times (2)$ avoids this.
- Dividing by $ac - b^2$ without saying that the question guarantees it is non-zero.
- Concluding $a = b = c$ from $\mathrm{Q}(a) = 0$ without using that a, b, c are real: the sum-of-squares argument fails for complex numbers.

Beyond STEP

The condition for two polynomials to have a common root is given in general by the *resultant*, a determinant built from their coefficients; the relation $(ac - b^2)(ab - c^2) = (bc - a^2)^2$ is exactly the resultant of these two quadratics set to zero. The expression $a^3 + b^3 + c^3 - 3abc$ is also the determinant of a circulant matrix, and it factorises completely as $(a + b + c)(a + \omega b + \omega^2 c)(a + \omega^2 b + \omega c)$ with ω a complex cube root of unity. For more on polynomials and their roots, try Allenby, *Rings, Fields and Groups*.

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